

ECON 101—Introduction to Microeconomics
Fall 2024

Midterm II

October 31, 2024

Part I: Multiple Choice

Please record all answers at the end of this section. Each question is worth 5 points for a total of 50 points.

Version	I	II	III	IV
Question				
1	(c)	(a)	(b)	(d)
2	(b)	(b)	(b)	(d)
3	(b)	(a)	(d)	(a)
4	(c)	(c)	(d)	(b)
5	(d)	(c)	(c)	(c)
6	(a)	(b)	(c)	(d)
7	(d)	(a)	(c)	(c)
8	(b)	(d)	(d)	(a)
9	(d)	(d)	(b)	(d)
10	(b)	(d)	(b)	(d)

End of the MCQ section. Please record all answers here!

	A	B	C	D
Question 1	☐	☐	☐	☐
Question 2	☐	☐	☐	☐
Question 3	☐	☐	☐	☐
Question 4	☐	☐	☐	☐
Question 5	☐	☐	☐	☐
Question 6	☐	☐	☐	☐
Question 7	☐	☐	☐	☐
Question 8	☐	☐	☐	☐
Question 9	☐	☐	☐	☐
Question 10	☐	☐	☐	☐

Part II: True/False Questions

For each of the following statements, indicate whether it is True or False, and provide a justification for your answer. Points will be awarded based on your explanation, not simply on choosing True or False. Without a justification, even a correct True/False answer will receive no points. Please write your answer only in the space provided. Each item is worth 10 points, for a total of 40.

1. Medina consumes only two goods, x_1 , and x_2 . When the price of x_1 increases, she consumes less of good x_2 .
Claim: x_2 can't be a Giffen good.
2. Let F denote food from GreenFood, and C denote consumption on everything else. Suppose that the typical NU student has a yearly budget of ₹400,000; the typical student spends ₹40,000 on F and ₹360,000 on C . NU's administration is planning to assist students by considering giving either an extra stipend in cash of ₹80,000 or giving students coupons for GreenFood worth ₹80,000.
 - (a) If F is a normal good, then a student will be strictly better off with ₹80,000 in cash than with ₹80,000 of coupons.
 - (b) If F is an inferior good, then a student will be strictly better off with ₹80,000 in cash than with ₹80,000 of coupons.

3. Rakhim consumes only sodas x_1 and pizza x_2 . Suppose that his preferences can be represented by $U(x_1, x_2) = \min\{x_1, x_2\}$, so sodas and pizza are perfect complements: Rakhim wants to drink one soda for every pizza he eats. If the price of soda increases, Rakhim's demand for both sodas and pizzas will decrease, and all change in the demand will be due to the substitution effect.

Part III: Long Problem on Consumer Theory

Answer each question only in the space provided. Be organized, clear, and concise. Points will be deducted for lack of organization or if your answer is unclear. The point value for each item is indicated in parentheses. This section is worth a total of 56 points.

Nazerke likes eating sweet pastries s and drinking coffee c for breakfast. Her preferences can be represented by the utility function

$$U(c, s) = c^3 s^2.$$

For your reference $MU_s = 2c^3s$, and $MU_c = 3c^2s^2$.

1. (4 points) Graph Nazerke's indifference curve corresponding to utility level $U = 64$. Coffee should be on the horizontal axis.
2. Suppose that Nazerke budgets m tenge for breakfast every week, and the price of coffee and sweet pastries are p_c and p_s respectively.
 - (a) (4 points) Write down the problem that Nazerke solves when she is deciding how many coffees and sweet pastries to consume each week. You are only required to write the problem, not solve it!

(b) (4 points) Solve for how many coffee and pastries Nazerke would consume as a function of p_c , p_s , and m .

(c) (4 points) Are coffee and pastries normal goods or inferior goods? Why?

3. Now suppose that Nazerke budgets ₹4500 for breakfast every day and only eats at Daily Cup, where the price of a pastry is ₹1000 and the price of a cup of coffee is ₹900. However, because Nazerke is part of the loyalty program at Daily cup, she receives a 50% discount on coffee.

(a) (4 points) Graph Nazerke's budget with coffee in the horizontal axis.

(b) (4 points) Find the number of coffee and pastries that Nazerke chooses to buy.

4. After a while, Daily cup decides to cancel the previous loyalty program and changes it to a new one: now a customer only gets a 50% discount on the first three cups of coffee they buy in any given day.
- (a) (5 points) Graph Nazerke's new budget set. Coffee should go on the horizontal axis. Carefully label every important point on the graph.

- (b) (5 points) What quantity of coffee and pastries would Nazerke choose to consume given the new budget set? How does this answer compare with the previous result? Explain your answers.

5. Suppose Daily cup decides to cancel its loyalty program

- (a) (4 points) Find Nazerke's consumption of coffee and pastries without any discounts.

- (b) (6 points) Comparing your answer to the one you got in 3.(b): how much of the change in Nazerke's consumption is due to the substitution effect? How much is due to the income effect? Perform the calculations and also illustrate your findings with a graph.

6. Nazerke is disappointed that the loyalty program has been canceled at Daily Cup. She decides to go to Health Project, where coffee sells for ₸800 Tenge and pastries for ₸800. Assume that the quality of coffee and pastries is the same between Health Project and Daily Cup.

- (a) (4 points) How many coffees and pastries does Nazerke buy at Health Project?

- (b) (4 points) Nazerke only has time to go either Daily or HP before class. Which one does she go to? Why?

- (c) (4 points) Now suppose that Nazerke's utility function for coffee and pastries is given by $U(c, s) = \sqrt{c + s}$. Which coffee shop does Nazerke go and how much of each good does she consume at that coffee shop?

Part IV: Bonus question (4 points)



Above, behold the magnificent, enigmatic, and exceedingly elusive House & Garden Leopard (*Panthera panquekis domesticus*). In honor of the magnificent *Panthera panquekis domesticus*, let your creativity shine! Draw a flag or coat of arms, write a poem, or create anything that comes to mind to celebrate this splendid creature.